


Before we start, we make a remark.

Let K be a field. Then $K[x]$ is Noetherian

but not Artinian, because $K[x]$ is ED

and hence is a PID, and

$$(x) \supseteq (x^2) \supseteq (x^3) \supseteq \dots$$

is a nonstabilizing descending chain of ideals.

In general, a polynomial ring can be defined

over a commutative ring.

Def 1: Let R be a commutative ring.

$$\text{Define } R[x] := \left\{ \sum_{i=0}^n a_i x^i : a_i \in R, \forall 0 \leq i \leq n, n \in \mathbb{N} \right\}.$$

Then. $R[x]$ is a ring with respect to

addition :

$$\sum_{i=0}^n a_i x^i + \sum_{j=0}^m b_j x^j = \sum_{k=0}^{\max\{n,m\}} (a_k + b_k) x^k$$

multiplication :

$$\sum_{i=0}^n a_i x^i \cdot \sum_{j=0}^m b_j x^j = \sum_{k=0}^{m+n} \left(\sum_{i+j=k} a_i b_j \right) x^k$$

We note if R is a domain.

then $R[x]$ is a domain.

We may still define the degree of a

polynomial in the usual way by looking

at the highest power of x with nonzero coefficients

Lemma 2: Let R be a domain and

$f, g \in R[x]$. Then $\deg fg = \deg f + \deg g$

Corollary 3: Let R be a domain.

Then $f \in R[x]$ is a unit iff $\deg f = 0$.

But once R is not a domain.

degree does not behave well under multiplication.

eg. In $\mathbb{Z}_9[x]$. we have

$$(3x+3)^2 = 9x^2 + 18x + 9 = 0.$$

$$(1+3x)(1+6x) = 1 + 9x + 18x^2 = 1.$$

Our goal is to study $\mathbb{Z}[x]$.

The good thing is that $\mathbb{Z}$ is a domain

So basic notions like degree still behaves

well. But $(2, x)$ is an ideal in $\mathbb{Z}[x]$

that is not principal. So $\mathbb{Z}[x]$ is not

a PID, although $\mathbb{Z}$ is. This implies

$\mathbb{Z}[x]$ is not ED. (why?) But one can

still ask: Is $\mathbb{Z}[x]$

1. Noetherian 2. UFD.

3. prime/irreducible elements

4. prime/maximal ideals.

To see why $\mathbb{Z}[x]$ is not ED with respect

to domain, note $a = 2x + 1$, $b = 3x + 1 \neq 0$

Then, if $a = bq + r$ for $q, r \in \mathbb{Z}[x]$,

with $r = 0$ or $\deg r < \deg b = \deg a$

we must have $\deg bq = \deg a$

which means $\deg q = 0$ i.e. $q \in \mathbb{Z}$

Moreover $LT(a) = LT(bq + r) = LT(bq)$

Clearly no integer $q \in \mathbb{Z}$ s.t.

$$2 = 3q.$$

Go back to the proof of $K[x]$ is ED

and see which step does not make sense

when R is a domain not a field

Question: Is $3x-3$ irreducible in $\mathbb{Z}[x]$?

Before we say anything concrete about $\mathbb{Z}[x]$.

We look at the larger polynomial rings

that contains them, i.e.

$$\mathbb{Z}[x] \subseteq \mathbb{Q}[x] \subseteq \mathbb{R}[x] \subseteq \mathbb{C}[x].$$

One important guiding principle is to

go to a larger place for help where things

work out nicely and deduce back.

eg. $x^2 - 2$ is irreducible in $\mathbb{Q}[x]$.

This should remind you of the example

that explains $x^2 + 1$ is irreducible in $\mathbb{R}[x]$.

Recall both $\mathbb{Q}[x]$, $\mathbb{R}[x]$ are polynomial rings over a field, and hence.

∴ $\mathbb{Q}[x]$, $\mathbb{R}[x]$ are UFD

2. Any degree 1 polynomial in $\mathbb{Q}[x]$
(or $\mathbb{R}[x]$)
irreducible.

3. Any degree 2 polynomial in $\mathbb{Q}[x]$
(or $\mathbb{R}[x]$)
either has a root in $\mathbb{Q}$ (or in $\mathbb{R}$)
or is irreducible (see homework)

So we conclude $x^2 - 2 = (x - \sqrt{2})(x + \sqrt{2})$,

is the unique irreducible factorization

in $\mathbb{R}[x]$. Suppose $x^2 - 2$ is not irreducible

in $\mathbb{Q}[x]$. i.e. it admits an irreducible

factorization in $\mathbb{Q}[x]$. But any

irreducible factorization in $\mathbb{Q}[x]$ is

also an irreducible factorization in $\mathbb{R}[x]$

which is uniquely $(x - \sqrt{2})(x + \sqrt{2})$.

Yet $x - \sqrt{2}, x + \sqrt{2} \notin \mathbb{Q}[x]$ as $\sqrt{2} \notin \mathbb{Q}$.

So $x^2 - 2$ is irreducible in $\mathbb{Q}[x]$.

eg. $3x+3$ is reducible in $\mathbb{Z}[x]$.

because. $3x+3 = 3(x+3)$

Note (1) 3 is nonzero, nonunit in $\mathbb{Z}[x]$

(2) 3 is irreducible in $\mathbb{Z}[x]$

because if $3 = fg$, then $\deg f = \deg g = 0$

i.e. $f, g \in \mathbb{Z}$, but 3 is prime in $\mathbb{Z}$.

(3) $x+3$ is irreducible in $\mathbb{Z}[x]$

because if $x+3 = fg$, then WLOG.

assume $\deg g = 0$ i.e. $g \in \mathbb{Z}$. Moreover

$LC(x+3) = LC(f) LC(g)$ which is $1 = LC(f) \cdot g$

so $g = \pm 1$ which is a unit in $\mathbb{Z}[x]$.